Where do NYC yellow cabs lose time, and did congestion pricing change it?

FDE Data Foundations - Track B (NYC TLC trip records). Jan 2024 (baseline) vs Jan 2025 (first month of the Manhattan congestion toll). Repository: README, code, source map, data model and evidence report. Every number below is produced by the pipeline (python -m pipeline run).

1. Which problem deserves to be solved

A client with trip data can ask many questions. We scored the candidates on how many trips they affect, whether a named owner can act on the answer, and whether the data can support the answer. The problem is chosen before any tooling.

Candidate problem	Evidence of size (Jan 2025)	Who acts on it	Can the data answer it?	Verdict
A. Fleet time lost in the Manhattan core	80% of trips are core-to-core; core weekday speed is 7.6 mph at midday vs 12.0 overnight	Fleet operations (shift and vehicle-hour planning)	Yes: time, distance and zones on every trip	**[CHOSEN]** largest and most actionable
B. Effect of the congestion toll	66% of trips charged the toll in its first month	TLC policy team	Partly: before/after only, no control group	**[CHOSEN]** as the intervention inside A
C. Airport trip reliability	8.3% of trips, but the longest (core->JFK 53 min)	Dispatch / airport queue	Yes	Tracked as metric M5
D. Fare and surcharge accuracy	32% of rows: total != sum of parts	TPEP vendors	No: vendor encoding, not documented	**[OUT OF SCOPE]** flag to data owner

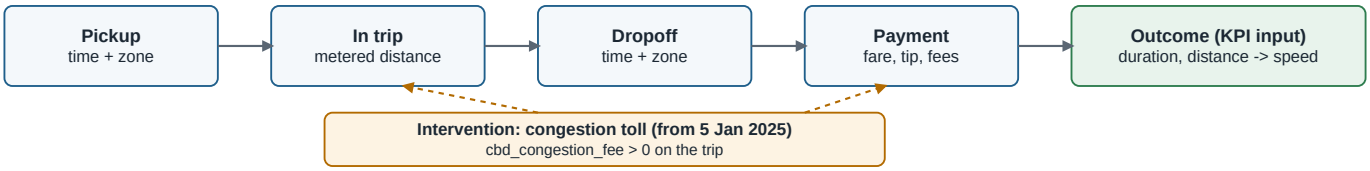
Problem statement. Fleet operations cannot tell with confidence where and when cabs are slowest in the core, or whether the toll made trips faster. A naive average over the raw files is distorted by stray timestamps, zero distances, voids and unknown zones. **Project KPI:** median weekday 07-19h speed (mph) of trips that start and end in the Manhattan core (TLC 'Yellow Zone'). **Decision supported:** where to put fleet hours, and whether to expect the toll to raise trips per hour.

2. Sources: from business question to system

Question	Source (owner)	Grain	Retrieval mode	Completeness evidence
Speed, duration, where, when	Yellow trip records Parquet (TLC / TPEP vendors)	trip	File over HTTPS	bytes = Content-Length; loaded rows = Parquet footer; sha256 + ETag kept
What counts as core / airport	Taxi zone lookup CSV (TLC)	zone	File (CSV)	265 rows; cross-checked with the API
Is the zone list right?	NYC Taxi Zones, Open Data (TLC)	zone shape	API (Socrata JSON)	paged rows = server count(*)
Do our totals match the regulator?	TLC Industry Indicators, Open Data	month x fleet	API (Socrata JSON)	paged rows = server count(*)
Validate, model, aggregate	DuckDB warehouse	-	SQL	run log per stage; month-level transactions

Workflow model. Each trip is one run of the workflow below. The toll is an intervention applied at payment; its expected effect is on in-trip time.

Entities: zone (263 + 2 unknown), vendor/TPEP provider, rate code, payment type. Grain: one row = one metered trip.



3. Is the data trustworthy? Profiling, validation and reconciliation

Profiling the raw files drove 10 **exclude** rules and 5 **warn** rules. No row is fixed or deleted silently: every raw row keeps one flag per rule and its first exclusion reason.

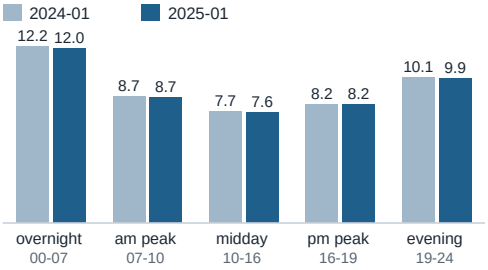
Rule (severity)	Why it matters for the KPI	2024-01 flagged / excluded	2025-01 flagged / excluded
R01 pickup outside file month (exclude)	stray 2002/2009 clocks leak into the wrong month	0.00% / 18	0.00% / 22
R02-R03 duration <= 0, < 1 min or > 3 h (exclude)	impossible, meter test or meter left on	1.25% / 37,104	1.19% / 41,488
R04-R06 zero / > 100 mi distance, > 65 mph (exclude)	no journey to measure, or odometer glitch	2.08% / 36,408	2.65% / 64,758
R07 zone 264/265 unknown (exclude)	cannot place the trip in core / non-core	1.06% / 27,882	0.82% / 25,839
R09 negative fare/total (exclude)	voids and refunds, not trips	1.27% / 31,678	4.16% / 126,427
W01 null passenger count + rate code (warn, kept)	see judgement call below	4.73%	15.54%
W04 total != sum of components (warn, kept)	vendor encoding; no fare-part metric is used	25.70%	31.99%

Independent check (API)	2024-01	2025-01	Gate
Our in-month trips/day vs TLC published	95,632 vs 94,742 (+0.9%)	112,103 vs 110,654 (+1.3%)	G3 pass (+/-5%)
Our mean minutes/trip vs TLC published	14.9 vs 14.6 (+2.2%)	14.6 vs 14.2 (+3.1%)	G4 pass (+/-10%)
Valid rows / days with trips	95.5% / 31 of 31	92.6% / 31 of 31	G1 (>= 85%), G2 pass

4. Metrics and what they show

Metric	2024-01	2025-01	Change
M3 KPI: core weekday 07-19h median speed	8.01 mph	7.98 mph	**-0.4%**
M1 valid trip rate	95.5%	92.6%	-3.0 pts
M2 valid trips per day	91,340	103,764	+13.6%
M4 median / p90 trip duration	11.7 / 28.7 min	11.7 / 27.5 min	p90 -1.2 min
M5 core -> JFK median duration (weekday day)	55.8 min	52.6 min	-3.2 min
M6 trips charged the congestion toll	n/a (no toll)	66.0%	-

Core weekday median speed by time band (mph)



Conclusion. Core speed did not change (8.01 -> 7.98 mph, -0.4%, n = 1,291,965 trips), even though 66% of trips paid the toll. The gains are on long trips: p90 duration -1.2 min and core->JFK -3.2 min. Midday (10-16h) is the slowest band in both years (7.6 mph).
Recommendation: do not plan for more trips per hour in the core because of the toll; plan midday core capacity around ~7.6 mph and treat airport runs as the segment that improved.

5. The key FDE judgement call: keep the rows with missing attributes

15.5% of Jan 2025 rows (4.7% in Jan 2024) have null passenger count and rate code (`payment_type = 0`). A generic 'drop incomplete rows' step would remove them. We kept them as a warning because: (1) their time, distance and zones are populated and plausible; (2) TLC counts them, so dropping them would move our trips/day from +1.3% to -14.4% against TLC's published figure; (3) the KPI barely moves either way (7.94 mph without them vs 7.98 with them). Deleting them would under-report demand for no gain in accuracy.

6. Facts, assumptions, unknowns and limitations

Type	Statement	Basis
[FACT]	Both monthly files are fully loaded and cover every day of the month.	Parquet footer row count; gate G2
[FACT]	Our volume and duration agree with TLC's published indicators within 1.3% and 3.1%.	Industry Indicators API
[FACT]	The zones API repeats IDs 56 and 103 and has no rows for 57, 104 and 105; the other 260 zones match.	Lookup CSV vs API cross-check
[ASSUMPTION]	'Manhattan core' = TLC Yellow Zone. This is close to, but larger than, the toll zone (south of 60th St).	TLC zone lookup
[ASSUMPTION]	Trips < 1 min, > 3 h, > 100 mi or > 65 mph are not real passenger trips.	Thresholds in config/pipeline.toml
[UNKNOWN]	What the payment_type 0 rows are ('Flex Fare' per the 2025 dictionary) and why they grew about 4x.	Needs TLC / vendor answer
[UNKNOWN]	Whether VendorID 1 folds surcharges into 'extra'; and TLC's method for its indicators.	Pattern only; no documentation
[LIMITATION]	Before/after comparison, not a controlled test: weather, events and fares also changed. Yellow cabs only (no Uber/Lyft). Zone-level location only.	Scope of the data

7. How the output is made repeatable

Reruns are idempotent: each month is replaced inside one transaction, and unchanged raw files (same ETag and sha256) are reused. If TLC re-publishes a file, the change is detected and it is downloaded again. HTTP calls retry with backoff. A month that is not yet published, or that fails, is logged and rolled back while other months continue. Fail gates block publishing (the metric status becomes 'blocked'). Every stage writes status, rows and seconds to a run log, and every metric row carries its run_id. The exit code is non-zero unless every month published. 5 automated tests cover the rules, reruns, gates and rollback.